

Network Analyses of Changing Character Relationships across the Alice in Wonderland Series

Elisabeth Giryes

Universität Zürich, Computational Linguistics, Zürich, Switzerland
elisabeth.giryes@uzh.ch

Jessica Karson

Universität Zürich, Digital Linguistics, Zürich, Switzerland
jessicaacton.karson@uzh.ch

July 1, 2024

Abstract

Lewis Carroll and the Wonderland world he created in his series including *Alice’s Adventures in Wonderland*, *Through the Looking-Glass*, *And What Alice Found There*, and *The Nursery Alice* are beloved for their nonsensical events and the antics of their characters. The complexity of the character relationships throughout this series can be analyzed and visualized through a series of network analyses. This paper presents the building of and subsequent reflection on this series of network visualizations. The creation of these network relationships relies on information extracted from the original texts which is undertaken by pre-processing (such as tokenizing by sentence and named entity recognition), structuring the network (with normalized relationship measures), and later visualization (with tools such as NetworkX (Hagberg et al. [2008 - 2024] and Gephi). These network findings are then independently interpreted for patterns, clusters, and changes over time and subsequently compared to a close-reading of the original series of novels. This Wonderland series is a particularly meaningful candidate for analysis because of its varying range of intended audiences across the books. *The Nursery Alice* is a simplified version of *Alice’s Adventures in Wonderland* created for younger readers (Clark [1986]). Furthermore, *Through the Looking-Glass*, *And What Alice Found There* takes place after the events of *Alice’s Adventures in Wonderland* and can be seen as a “coming-of-age tale” intended for continuing readers and teens

(Coster [2016]). This variation allows this project to study how character relationships change across books with a range of intended audiences using the medium of networks visualizations created from the source novels. Through this series of network analyses of character relationships extracted from across the Alice series, this project identifies the presence of a change in the complexity of the relationships over time as the audience is expected to grow along with the Alice character and be able to understand increasingly complex character dynamics.

Keywords: Network Analysis; Network Theory; Literature; Computational Linguistics; Natural Language Processing; English Literature

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1 Introduction

This report introduces the Alice in Wonderland series and its three novels that include the central events, characters, and happenings of the Wonderland Universe. The aptness of this

series for evaluation using network analysis is then revealed and the concept of network analysis is summarized. The research methodology used to build these particular networks using the tools of NetworkX, Gephi, and Flourish is then outlined in section 2. The results of these analyses are then shown visually (as is essential to the evaluation of this type of literary analysis). Importantly, the discussion section of this paper then details the structures of these networks and involves interpretations, comparisons, and case studies of important character relationships. A close-reading comparison to the results of these networks is also discussed bearing in mind the caveat that individual readers may have varying perspectives and perceptions of character relationships. The idea that this distant-reading approach of network analysis may enrich or alleviate these personal biases found through close-reading is also touched upon. This area of literary analysis has great potential for expansion, and, on a smaller scale, this project has possibilities for future work which will be covered before conclusions are offered.

Character relationships are central to our understanding of the texts we consume as readers and as computational researchers. As Chaturvedi et al. writes, "*studying characters plays a vital role in computationally representing and interpreting narratives*," (Chaturvedi et al. [2016]). By viewing the Alice in Wonderland series principally through its character relationships, we are able to abstract the text into a concentrated understanding of the happenings of the novel and gain new perspective on the classic and well-loved series.

1.1 The Alice in Wonderland Series and Its Author

Nineteenth-century British author Lewis Carroll (born Charles Lutwidge Dodgson), wrote extensively throughout his life. Though probably most famous for *Alice's Adventures in Wonderland* (1865) and *Through the Looking-Glass, And What Alice Found There* (1872), Carroll also wrote literary pieces for the journal *The Train* and other publications, as well as, mathematics papers ("A Syllabus of Plane Algebraical Geometry" (1860), "The Formulae of Plane Trigonometry" (1861)), poems (such as "The Path of Roses" (1856)), and other works, (Poetry Foundation [2024]). One piece he wrote for *The Train* called "*Upon the Lonely Moor*" actually was the basis for the later White Knight's ballad in *Alice's Adventures in Wonderland* (1865), (Poetry Foundation [2024]).

The characters within *Alice's Adventures in Wonderland* often refer to the personal relationships author Carroll had in real life. Poetry foundation writes that [*Alice's Adventures in Wonderland*] makes bold references to the practices and politics of the day, and mentions specific friends and acquaintances of the author—not always in a complimentary fashion, (Poetry Foundation [2024]). Roger Lancelyn Green in his edits of collections of Carroll's personal letters, supposed that Lorina and Edith, sisters of Alice Liddell, are the inspirations for the characters Lory and the Eagle and that the two younger sisters, Rhoda and Violet, are the characters Rose and the Violet (Green [1953]). Furthermore, one of the most iconic characters to come out of the Alice series, the Mad Hatter, Green thought to

be Theophilus Carter, who ran a shop in Oxford (Green [1953]).

This intermixing of fictional character relationships with real individuals in Carroll's life show the significance and noteworthiness of character dynamics within the Alice series. The amount that can be revealed about a story through the study of its character relationships through this projects series of network analyses will be shown in this report.

The suitability of this Alice series for this network analysis is threefold.

Firstly, the popularity and beloved status of this series is evident. Warren Weaver found 47 language translations of *Alice's adventures in Wonderland* already by 1963 (Weaver [1964]). Poetry Foundation writes that the series popularity can be attributed in part to "the accessibility of the Alice books, both in terms of language and the fact that they have never been out of print," (Poetry Foundation [2024]).

Secondly, due to the publishing of this series in the 1800s, the copyright status of this series allows it to be accessible through Project Gutenberg in the public domain (Gutenberg [2024]).

Finally, and importantly, the series of three novels assessed in this project have a special relationship to one another. *Alice's Adventures in Wonderland* (1865) is a story centering around a young girl named Alice and her journey in a new fantasy world she discovers after falling down a rabbit hole. Then, *Through the Looking-Glass, And What Alice Found There* (1872) follows Alice's continued adventures which take place after the events of *Alice's Adventures in Wonderland*. In *Through the Looking-Glass, And What Alice Found There* Alice can be seen as going through a coming-of-age arc from child to adult, intended for continuing readers (Coster [2016]). Finally assessed in this analysis, *The Nursery Alice* (1889), is an adapted version of the original *Alice's Adventures in Wonderland*. It is a simplified version of the same story that was targeted for younger children and is "intended to make Alice in Wonderland more accessible to the young" (Poetry Foundation [2024]). This range of (1) a novel intended for young audiences, (2) middle-grade audiences, and (3) teenage audiences, allows us to study the varying complexity of character relationships present across the novels through network analyses for each novel. In this way, this project can track and identify the presence of a change in the intricacy of the relationships over time as the audience is expected to grow along with the Alice character and be able to understand increasingly complex character dynamics.

1.2 What is Network Analysis?

Network analysis is based upon network theory which studies the connections between large groups of objects and is written about extensively by Franco Moretti in his "Network Theory, Plot Analysis" article (Moretti [2011]). A network is made up of vertices or points and edges or lines between points. Weighting and thickness of edges can represent stronger connections between the vertices they connect. The closeness or shortness of the edges can also show a more important connection between the vertices. For the purposes of this

project, each vertex represents a character in the Alice series and each edge is a connection between those two characters.

Network analysis can be performed based on different criteria. For example, it can be undertaken based on proximity between characters in a text. For example, if two characters appear within two sentences of each other in a passage, this would count positively towards their connection. Another criteria could be a specific interaction between characters. For example, if two characters have some sort of interchange, like they have spoken words to each other in the text, this may also be used as a criteria for a closer network connection.

Network analysis offers a distant-reading approach to evaluating a text. This distant approach allows many benefits that can reveal key connections and insights about a text. A benefit of network analysis is that we may compress time within a novel. We are essentially able to view all of the character interactions of the novel at once in a compressed network view. Thus, instead of what occurs later in the novel taking precedence in the mind of the reader, all of the happenings of the novel can be considered at once (Moretti [2011]). Additionally, network analysis allows us to isolate certain regions of the novel within the network. For example, the main character of a series can be removed to analyze what the side and surrounding character interactions look like. Generally, network analysis enables evaluators to view a novel in an abstract way as a model rather than a string of words (Moretti [2011]).

Of course, in order to perform this analysis, the necessary network building with a computation model must occur. There are a few available tools for this task. Both NetworkX and Gephi are two such tools.

NetworkX was chosen as one of our tool options in this project for its status as an accessible Python package (open source 3-clause BSD license) (Hagberg et al. [2008 - 2024]). Its purpose-built standard graph algorithms align perfectly with the aim of this project and its data structures allow for graphs, digraphs, and multigraphs. Also, nodes with NetworkX "can be "anything" (e.g., text, images, XML records)" and "edges can hold arbitrary data (e.g., weights, time-series)" (Hagberg et al. [2008 - 2024]). These features allow for a flexible and streamlined network creation.

Gephi is also a suitable visualization option. Work with this model was attempted principally to compare our NetworkX results to another network building tool (Bastian [2022]). This tool is also available publicly through its GNU General Public License (GPL) v3. This comparison will be further discussed in section 2.5

2 Methodology

The following process was followed in undertaking this network building project.

2.1 Preparation

In preparation to begin this project, we completed background research on topics covered in our "Text Analytics in the Digital Humanities" course and found our interest in network theory. We then researched topic-related papers and possible network-building tools such as NetworkX, Gephi and guides and tutorials on how these programs and tools commonly function. We decided to process our texts with Python. We also narrowed our choices of series of novels and finally chose our the Alice series for the reasons described in the previous 1.1 section.

2.2 Data Collection and Pre-processing

With our chosen text, we pre-processed the data that was accessed through downloaded text files from Project Gutenberg (Gutenberg [2024]). This pre-processing involved removing the header and footer that is almost always included in files downloaded from Project Gutenberg. These sections include historical and background information, a table of contents, later citation information, and other information irrelevant to our analysis of only the actions and interactions of the fictional Alice stories.

2.3 Data Processing

Once only the base Alice texts were left, we separated the text into sentence units. We then used spaCy to perform Named Entity Recognition (NER) using its base English model (ExplosionAI [2024]). Although characters in the Alice series often have unconventional names such as animal names and number, spaCy still performed satisfactorily as a first step in name recognition. In order to ensure full name collection, we used a JSON file of all the names that appear in the Alice in Wonderland series to find and filter character names. This file also included all versions of each character name so that the *Hatter* and the *Mad Hatter*, for example were both captured and understood to be the same individual. This list was collected from SparkNotes compiled character lists (SparkNotes [2024b]) (SparkNotes [2024a]).

2.4 Relationship Extraction

In order to extract relationships we built the ‘process_book’ script. The following is a step-by-step guide of what the ‘process_book’ script does:

The script begins by importing the necessary libraries, including `json`, `re`, `pandas`, and `spacy`. It then loads the spaCy English model `en_core_web_sm` for performing Named Entity Recognition (NER).

A function named `perform_ner` is defined to use spaCy for extracting named entities from a given document. The main function, `process_book`, is then defined to process the

book and analyze character relationships and frequencies.

This main function starts by loading the original text from the specified file and converting it to lowercase for uniformity. It also reads the character data from a provided JSON file, creating a list of characters and their aliases. If a specific character needs to be removed from the analysis, it is excluded from this list.

The text is tokenized into sentences using a regular expression, and a helper function `filter_entity` is defined to filter out non-character entities from each sentence and handle cases where characters have a "the"-prefix.

For each sentence, the script processes the text to extract character entities and context entities, storing each sentence along with its entities and context entities in a list of dictionaries. This list is then converted into a pandas DataFrame.

Next, sentences that do not have at least two unique character entities in their context are filtered out. A dictionary is initialized to store counts of character relationships. The script then iterates through the filtered sentences, creating pairs of character relationships based on context entities and incrementing the count for each relationship pair in the dictionary.

The relationship data is converted into a pandas DataFrame, where the relationship pairs are split into separate columns (`source` and `target`). Counts for relationships appearing in both directions are aggregated, normalized by the total number of relationships, and saved as a CSV file with a specified output prefix.

Character mentions and frequencies are calculated by finding mentions of each character in the text using regular expressions. The frequency of each character is computed as a proportion of total mentions, and this frequency data is converted into a pandas DataFrame, sorted by frequency, and saved as a CSV file.

This process is repeated to analyze each book with and without a specific character removed, enabling comparison of character relationships and frequencies with and without the influence of that character.

2.5 Visualize Network

To visualize the relationships and co-occurrences of characters within each novel, we used the co-occurrence scores data frames, which we exported as CSV files.

These CSV files served as input for the NetworkX package to create visualizations of the character relationship networks. To begin with, we imported the necessary libraries: pandas for data manipulation, NetworkX for creating and managing complex networks, and Matplotlib for plotting the graphs. We loaded the relationship data from the CSV file into a pandas DataFrame and then used this DataFrame to create a graph in NetworkX. The relationships between characters were represented as edges in the graph, with the co-occurrence counts used as edge attributes.

For the initial visualization, we created a simple plot using Matplotlib. We set the figure size and defined a layout for the nodes using the spring layout algorithm, which positions

the nodes in a way that simulates a force-directed layout, making the visualization more readable. The graph was then drawn with node labels, node colors, and an edge color map. We also included edge labels to display the normalized counts of character co-occurrences.

While this initial visualization provided a good overview of the character relationships, it had limitations. The edges were not weighted, which made it difficult to distinguish the strength of the relationships, and the visualization could become cluttered with numerous overlapping edges.

To enhance our analysis, we used the same CSV files as inputs for Gephi, a powerful network analysis and visualization software. In Gephi, we applied the Force Atlas 2 layout, which further improved the readability of the graph by distributing the nodes based on the forces between them. We also ranked the nodes by degree, which highlighted the most connected characters, and ranked the edges by normalized count, providing a clearer view of the strength of relationships. Gephi’s interactive capabilities allowed us to explore the network in greater detail and offered various options for visual customization.

Finally, we imported the CSV data into Flourish Studio, an easy-to-use tool for creating interactive visualizations. Flourish Studio’s 3D interactive tool allowed us to create engaging and shareable visualizations with minimal effort. It automatically handled many of the visualization tasks, making it a convenient option for quickly generating and publishing interactive network graphs. The interactive visualization is public and accessible under this [Link](#).

By processing each novel individually, we were able to create separate visualizations for each book. This approach allowed us to compare the character relationships and interactions across different novels, providing deeper insights into the narrative structure and character dynamics.

2.6 Network Evaluation

In order to evaluate our created networks we visually assessed the structures and centrality of main characters. We also removed characters to observe the effects on the surrounding relationships. Furthermore, this distant-reading approach was combined with our close-reading of the text to evaluate how the network reflects a subjective reading of the text. These evaluations are detailed in the following sections 3 and 4.

3 Results

Character Frequencies

The set of character frequencies represents how often each character appears or is mentioned in the text, with the normalized values indicating the proportion of the total mentions that each character accounts for. This gives an overview of the significance of each character in

the Alice's in Wonderland series.

Table 1: *Alice's Adventures in Wonderland* Character Frequencies (normalized)

character	frequency
alice	0.4521
hatter	0.0639
gryphon	0.0628
mock turtle	0.0571
duchess	0.0457
two	0.0434
dormouse	0.0428
mouse	0.0400
march hare	0.0342
caterpillar	0.0320
white rabbit	0.0228
bill	0.0194
dodo	0.0143
cook	0.0143
pigeon	0.0126
five	0.0091
lory	0.0068
seven	0.0068
cheshire cat	0.0057
queen of hearts	0.0046
knave of hearts	0.0029
duck	0.0029
eaglet	0.0029
frog-footman	0.0011

Table 2: *Alice's Adventures in Wonderland* Character Frequencies without Alice (normalized). This is an example table showing the impact to frequencies when the main character is removed.

character	frequency
hatter	0.1167
gryphon	0.1146
mock turtle	0.1042
duchess	0.0833

character	frequency
two	0.0792
dormouse	0.0781
mouse	0.0729
march hare	0.0625
caterpillar	0.0583
white rabbit	0.0417
bill	0.0354
dodo	0.0260
cook	0.0260
pigeon	0.0229
five	0.0167
lory	0.0125
seven	0.0125
cheshire cat	0.0104
queen of hearts	0.0083
knave of hearts	0.0052
duck	0.0052
eaglet	0.0052
frog-footman	0.0021

Table 3: *Nursery Alice* Character Frequencies (normalized)

character	frequency
alice	0.5236
duchess	0.0707
caterpillar	0.0471
white rabbit	0.0419
march hare	0.0393
two	0.0314
hatter	0.0314
dodo	0.0288
cook	0.0262
mouse	0.0262
bill	0.0262
duck	0.0236
queen of hearts	0.0157
gryphon	0.0157

character	frequency
dormouse	0.0131
five	0.0105
mock turtle	0.0079
lory	0.0079
seven	0.0052
knave of hearts	0.0052
eaglet	0.0026

Table 4: *Through the Looking-Glass* Character Frequencies (normalized)

character	frequency
alice	0.5420
red queen	0.0667
humpty dumpty	0.0584
white queen	0.0417
tweedledum	0.0405
tweedledee	0.0310
unicorn	0.0262
sheep	0.0226
gnat	0.0214
lion	0.0208
lily	0.0191
tiger-lily	0.0143
rose	0.0143
hatta	0.0131
haigha	0.0119
white king	0.0071
frog	0.0071
white knight	0.0071
fawn	0.0071
red knight	0.0060
red king	0.0060
guard	0.0060
daisy	0.0024
daisies	0.0024
goat	0.0024
violet	0.0012

character	frequency
larkspur	0.0012

Character Relationship Data

The character frequencies in the dataset indicate how often each character (source) interacts or appears in relation to another character (target) in the text, with both raw counts and normalized counts reflecting the relative frequency of these interactions. This information helps to understand the significance and relationships of different characters in the Alice in Wonderland series.

Table 5: *Alice’s Adventures in Wonderland* Relationship Data

source	target	count	normalized_count
alice	bill	11	0.0144
alice	caterpillar	22	0.0287
alice	cheshire cat	10	0.0131
alice	cook	11	0.0144
alice	dodo	8	0.0104
alice	dormouse	27	0.0352
alice	duchess	38	0.0496
alice	duck	4	0.0052
alice	eaglet	4	0.0052
alice	five	6	0.0078
alice	frog-footman	1	0.0013
alice	gryphon	42	0.0548
alice	hatter	41	0.0535
alice	knave of hearts	3	0.0039
alice	lory	6	0.0078
alice	march hare	30	0.0392
alice	mock turtle	42	0.0548
alice	mouse	51	0.0666
alice	pigeon	8	0.0104
alice	queen of hearts	5	0.0065
alice	seven	5	0.0065
alice	two	48	0.0627
alice	white rabbit	22	0.0287
bill	duchess	1	0.0013
bill	mock turtle	1	0.0013

source	target	count	normalized_count
bill	mouse	1	0.0013
bill	two	2	0.0026
bill	white rabbit	1	0.0013
caterpillar	gryphon	1	0.0013
caterpillar	two	3	0.0039
caterpillar	white rabbit	1	0.0013
cheshire cat	cook	2	0.0026
cheshire cat	duchess	2	0.0026
cheshire cat	two	1	0.0013
cook	dormouse	1	0.0013
cook	duchess	7	0.0091
cook	five	1	0.0013
cook	hatter	1	0.0013
cook	mouse	1	0.0013
cook	seven	1	0.0013
cook	two	1	0.0013
cook	white rabbit	1	0.0013
dodo	duck	1	0.0013
dodo	eaglet	4	0.0052
dodo	lory	1	0.0013
dodo	mouse	2	0.0026
dodo	two	2	0.0026
dormouse	hatter	23	0.0300
dormouse	march hare	16	0.0209
dormouse	mouse	33	0.0431
dormouse	queen of hearts	1	0.0013
dormouse	two	6	0.0078
dormouse	white rabbit	1	0.0013
duchess	frog-footman	1	0.0013
duchess	gryphon	1	0.0013
duchess	hatter	1	0.0013
duchess	march hare	1	0.0013
duchess	mock turtle	2	0.0026
duchess	mouse	2	0.0026
duchess	white rabbit	3	0.0039
duck	eaglet	1	0.0013
duck	gryphon	1	0.0013
duck	lory	1	0.0013

source	target	count	normalized_count
duck	mouse	3	0.0039
eaglet	lory	1	0.0013
eaglet	mouse	1	0.0013
five	seven	5	0.0065
five	two	3	0.0039
gryphon	march hare	2	0.0026
gryphon	mock turtle	36	0.0470
gryphon	mouse	2	0.0026
gryphon	two	6	0.0078
gryphon	white rabbit	5	0.0065
hatter	knave of hearts	1	0.0013
hatter	march hare	27	0.0352
hatter	mouse	25	0.0326
hatter	queen of hearts	2	0.0026
hatter	two	6	0.0078
hatter	white rabbit	1	0.0013
bill	knave of hearts	1	0.0013
knave of hearts	queen of hearts	3	0.0039
knave of hearts	white rabbit	2	0.0026
lory	mouse	3	0.0039
march hare	mouse	13	0.0170
march hare	queen of hearts	1	0.0013
march hare	two	7	0.0091
march hare	white rabbit	1	0.0013
caterpillar	mock turtle	1	0.0013
duck	mock turtle	2	0.0026
mock turtle	two	5	0.0065
mouse	queen of hearts	1	0.0013
mouse	two	7	0.0091
mouse	white rabbit	2	0.0026
pigeon	two	1	0.0013
bill	queen of hearts	1	0.0013
gryphon	queen of hearts	1	0.0013
queen of hearts	white rabbit	3	0.0039
seven	two	3	0.0039
two	white rabbit	2	0.0026
mock turtle	white rabbit	1	0.0013

Table 6: *Nursery Alice* Relationship Data

source	target	count	normalized_count
alice	bill	3	0.0238
alice	caterpillar	6	0.0476
alice	cheshire cat	1	0.0079
alice	cook	3	0.0238
alice	dodo	3	0.0238
alice	dormouse	2	0.0159
alice	duchess	8	0.0635
alice	duck	1	0.0079
alice	eaglet	1	0.0079
alice	five	1	0.0079
alice	frog-footman	1	0.0079
alice	gryphon	2	0.0159
alice	hatter	3	0.0238
alice	knave of hearts	1	0.0079
alice	lory	1	0.0079
alice	march hare	4	0.0317
alice	mock turtle	3	0.0238
alice	mouse	10	0.0794
alice	pigeon	1	0.0079
alice	queen of hearts	1	0.0079
alice	seven	1	0.0079
alice	two	5	0.0397
alice	white rabbit	11	0.0873
bill	duchess	1	0.0079
bill	mock turtle	1	0.0079
bill	mouse	1	0.0079
bill	two	2	0.0159
bill	white rabbit	1	0.0079
caterpillar	gryphon	1	0.0079
caterpillar	two	2	0.0159
caterpillar	white rabbit	1	0.0079
cheshire cat	cook	1	0.0079
cheshire cat	duchess	1	0.0079
cheshire cat	two	1	0.0079
cook	dormouse	1	0.0079
cook	duchess	7	0.0519

source	target	count	normalized_count
cook	five	1	0.0079
cook	hatter	1	0.0079
cook	mouse	1	0.0079
cook	seven	1	0.0079
cook	two	1	0.0079
cook	white rabbit	1	0.0079
dodo	duck	1	0.0079
dodo	eaglet	2	0.0159
dodo	lory	1	0.0079
dodo	mouse	1	0.0079
dodo	two	1	0.0079
dormouse	hatter	3	0.0238
dormouse	march hare	3	0.0238
dormouse	mouse	4	0.0317
dormouse	queen of hearts	1	0.0079
dormouse	two	6	0.0476
dormouse	white rabbit	1	0.0079
duchess	frog-footman	1	0.0079
duchess	gryphon	2	0.0159
duchess	hatter	1	0.0079
duchess	march hare	1	0.0079
duchess	mock turtle	2	0.0159
duchess	mouse	2	0.0159
duchess	white rabbit	3	0.0238
duck	eaglet	1	0.0079
duck	gryphon	1	0.0079
duck	lory	1	0.0079
duck	mouse	3	0.0238
eaglet	lory	1	0.0079
eaglet	mouse	1	0.0079
five	seven	5	0.0397
five	two	3	0.0238
gryphon	march hare	2	0.0159
gryphon	mock turtle	36	0.2857
gryphon	mouse	2	0.0159
gryphon	two	6	0.0476
gryphon	white rabbit	5	0.0397
hatter	knave of hearts	1	0.0079

source	target	count	normalized_count
hatter	march hare	27	0.2143
hatter	mouse	25	0.1984
hatter	queen of hearts	2	0.0159
hatter	two	6	0.0476
hatter	white rabbit	1	0.0079
bill	knave of hearts	1	0.0079
knave of hearts	queen of hearts	3	0.0238
knave of hearts	white rabbit	2	0.0159
lory	mouse	1	0.0079
march hare	mouse	13	0.1032
march hare	queen of hearts	1	0.0079
march hare	two	7	0.0556
march hare	white rabbit	1	0.0079
caterpillar	mock turtle	1	0.0079
duck	mock turtle	2	0.0159
mock turtle	two	5	0.0397
mouse	queen of hearts	1	0.0079
mouse	two	7	0.0556
mouse	white rabbit	2	0.0159
pigeon	two	1	0.0079
bill	queen of hearts	1	0.0079
gryphon	queen of hearts	1	0.0079
queen of hearts	white rabbit	3	0.0238
seven	two	3	0.0238
two	white rabbit	2	0.0159
mock turtle	white rabbit	1	0.0079

Table 7: *Through the Looking Glass* Relationship Data

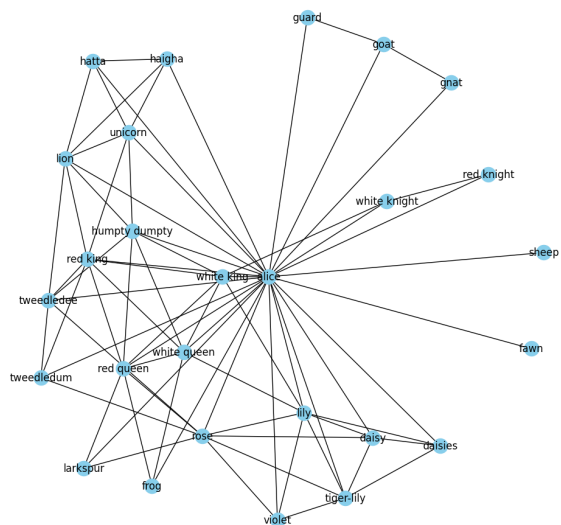
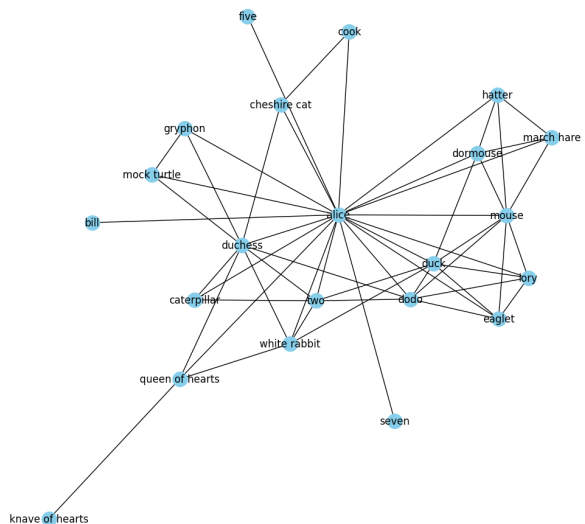
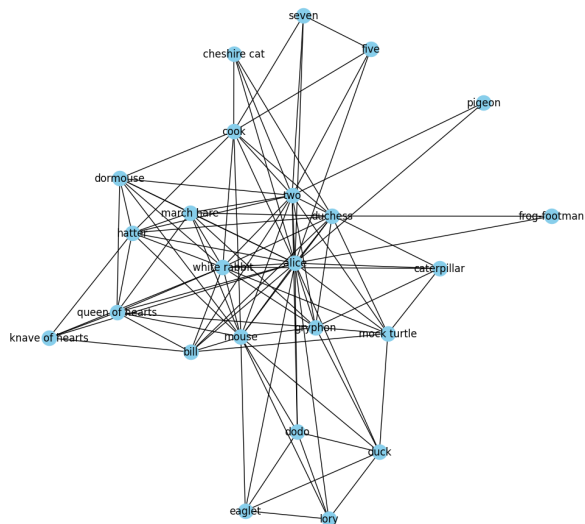
source	target	count	normalized_count
alice	daisies	3	0.0059
alice	daisy	1	0.0020
alice	fawn	6	0.0118
alice	frog	7	0.0138
alice	gnat	15	0.0295
alice	goat	3	0.0059
alice	guard	5	0.0098

source	target	count	normalized_count
alice	haigha	9	0.0177
alice	hatta	5	0.0098
alice	humpty dumpty	43	0.0846
alice	larkspur	1	0.0020
alice	lily	12	0.0236
alice	lion	14	0.0276
alice	red king	4	0.0079
alice	red knight	5	0.0098
alice	red queen	54	0.1063
alice	rose	10	0.0197
alice	sheep	20	0.0394
alice	tiger-lily	10	0.0197
alice	tweedledee	32	0.0630
alice	tweedledum	34	0.0669
alice	unicorn	16	0.0315
alice	violet	2	0.0039
alice	white king	10	0.0197
alice	white knight	8	0.0157
alice	white queen	39	0.0768
daisies	daisy	1	0.0020
daisies	lily	2	0.0039
daisies	tiger-lily	2	0.0039
daisy	rose	1	0.0020
gnat	goat	2	0.0039
goat	guard	1	0.0020
haigha	hatta	4	0.0079
haigha	lion	1	0.0020
haigha	unicorn	5	0.0098
hatta	lion	3	0.0059
hatta	unicorn	3	0.0059
humpty dumpty	tweedledee	1	0.0020
humpty dumpty	white king	1	0.0020
humpty dumpty	white queen	1	0.0020
larkspur	rose	1	0.0020
daisy	lily	1	0.0020
lily	rose	4	0.0079
lily	tiger-lily	10	0.0197
lily	violet	1	0.0020

source	target	count	normalized_count
lily	white queen	2	0.0039
humpty dumpty	lion	1	0.0020
lion	unicorn	13	0.0256
lion	red king	2	0.0039
red king	red queen	1	0.0020
red king	tweedledee	1	0.0020
red king	unicorn	1	0.0020
red king	white king	1	0.0020
red king	white queen	1	0.0020
red knight	white knight	4	0.0079
frog	red queen	1	0.0020
humpty dumpty	red queen	1	0.0020
larkspur	red queen	1	0.0020
red queen	rose	1	0.0020
red queen	white king	1	0.0020
red queen	white queen	24	0.0472
rose	tiger-lily	7	0.0138
rose	tweedledee	1	0.0020
rose	tweedledum	1	0.0020
rose	violet	1	0.0020
daisy	tiger-lily	1	0.0020
tiger-lily	violet	1	0.0020
lion	tweedledee	1	0.0020
tweedledee	tweedledum	20	0.0394
red king	tweedledum	1	0.0020
humpty dumpty	unicorn	1	0.0020
lily	white king	1	0.0020
white king	white knight	1	0.0020
white king	white queen	1	0.0020
frog	white queen	1	0.0020

3.1 Visualizations

From these tables of data, the following series of networks were created. The first set of three networks visualize the data with the NetworkX tool:



The following series of networks were created with Flourish and allow us to show the weighted relationships within the structure and provides an optional 3D interactive tool (interactive version found here: [Flourish](#)).

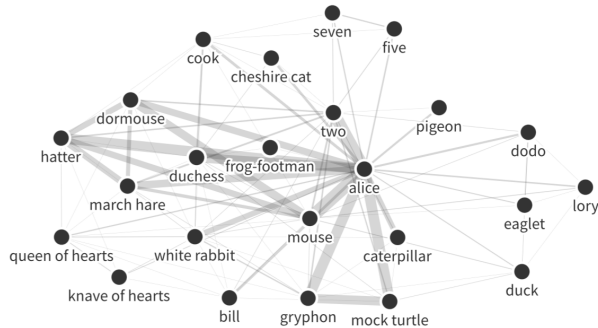


Figure 4: Flourish Visualization for *Alice's Adventures in Wonderland*

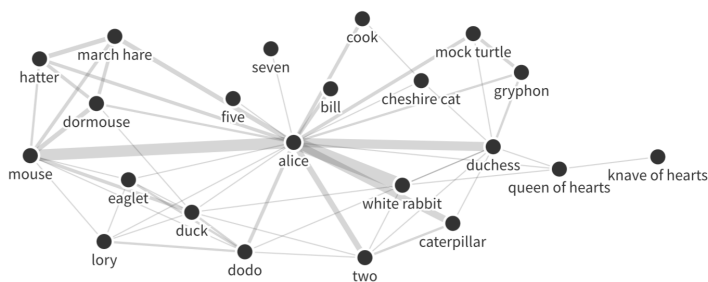


Figure 5: Flourish Visualization for *The Nursery Alice*

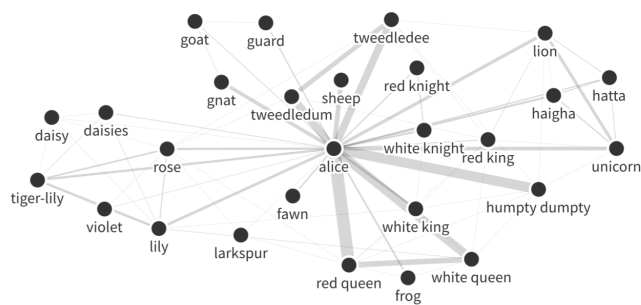


Figure 6: Flourish Visualization for *Through the Looking-Glass, And What Alice Found There*

The following series of networks were created with the Gephi network tool:

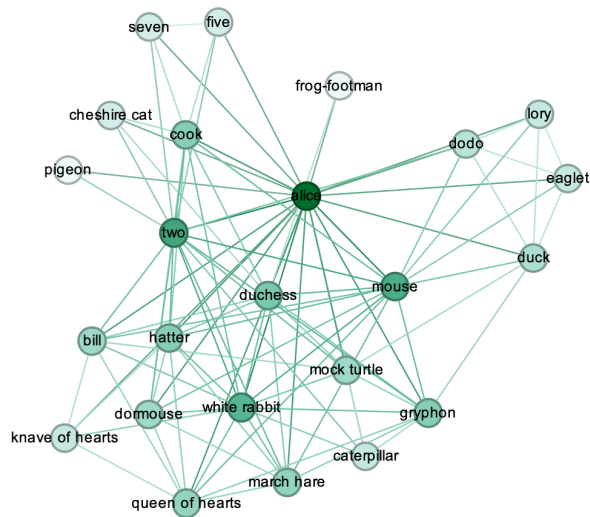


Figure 7: Gephi Visualization for *Alice's Adventures in Wonderland*

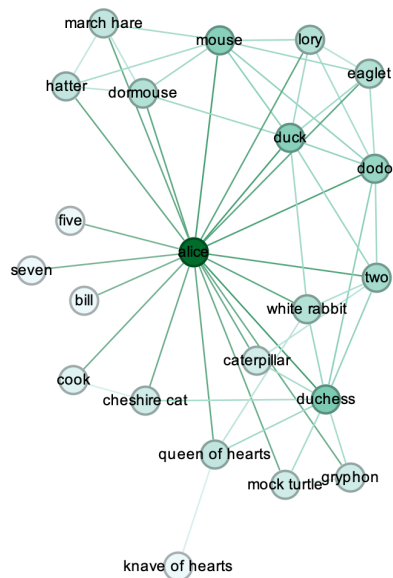


Figure 8: Gephi Visualization for *Nursery Alice*

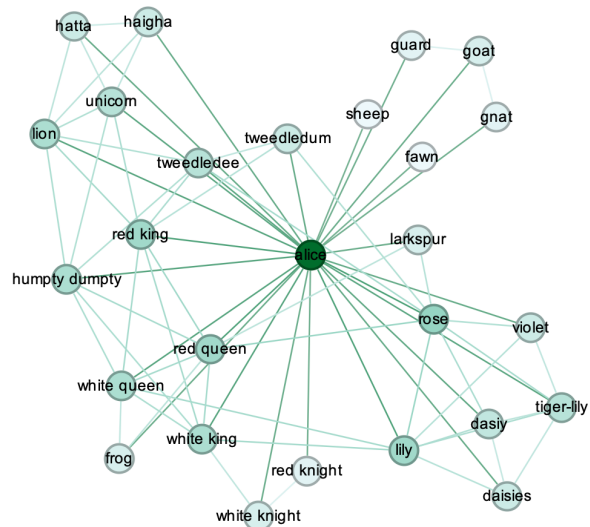


Figure 9: Gephi Visualization for *Through the Looking Glass*

4 Discussion and Analysis

4.1 Comparison Visualization Tools

For our visualizations, we used three different visualization tools: NetworkX, Flourish Studio and Gephi. We want to briefly compare these tools.

NetworkX Visualizations (Figures 1, 2, and 3) The nodes are represented as blue circles with character names displayed. Edges are simple straight lines connecting the nodes. The layout is somewhat random but attempts to cluster connected nodes together. It uses a force-directed algorithm to position nodes.

Typically, NetworkX visualizations are static when displayed in images. However, when used in an interactive environment (like a Jupyter Notebook), they can be interactive but generally lack advanced features like filtering or zooming. Our visualization appears somewhat cluttered, especially in denser parts of the network, making it hard to distinguish individual nodes and edges. And lastly, the edges are not weighted, thus implying that every connection is equal, which is not correct.

Flourish Studio Visualizations (Figures 4, 5, and 6) The nodes are black circles with character names adjacent. Edges are gray lines connecting nodes. This style is more minimalist compared to NetworkX.

The layout is more structured and spaced out, possibly using a similar force-directed algorithm but with better separation of clusters. The edges are weighted automatically.

Flourish Studio is known for its interactive visualizations, allowing for features such as hovering over nodes to get more information, zooming, and filtering, which are not visible in the static images but are inherent to the tool. The visualization is cleaner and more organized, reducing visual clutter and making it easier to see individual connections.

Gephi Visualization (Figures 7, 8, and 9) Green nodes with green lines for edges - the colour intensity is based on node degree. Labels are clearly attached to nodes. Nodes and edges are customizable in terms of color, size, and thickness. Gephi offers advanced layout algorithms like ForceAtlas2, which we used here. The nodes are well-distributed, with clear separation between clusters, leading to a less cluttered and more organized visualization. Gephi visualizations are highly interactive. Gephi allows for zooming, panning, and real-time filtering. One can adjust the visualization dynamically. Also including a high level of customization available for both nodes and edges. One can apply different colors, sizes, and other styles to better represent the data.

In conclusion, each tool has its strengths:

- **NetworkX** is suitable for basic, straightforward network visualizations and preliminary analysis.
- **Flourish Studio** offers polished, clean, and interactive visualizations, perfect for presentations and web-based explorations.

- **Gephi** excels in creating detailed, highly customizable, and interactive visualizations, ideal for complex network analysis.

4.2 Interpretation of Network Analysis Results

Character Complexity In *Alice's Adventures in Wonderland*, the character network is diverse and vibrant. Alice's interactions with various characters create a hub-and-spoke model, with her at the center. In *Nursery Alice*, the network is significantly simplified, reducing the number of characters and interactions. While strong connections remain with key characters such as the Mouse and the Duchess, relationships with others, like the Hatter, are weakened or omitted. In contrast, *Through the Looking Glass* presents a more intricate network, introducing new characters and complex interactions that reflect Alice's growth and the novel's advanced themes.

Interaction Patterns In *Alice's Adventures in Wonderland*, interactions are primarily episodic, with Alice moving from one character encounter to another, focusing on her perspective. *Nursery Alice* simplifies these interactions, concentrating on a few central figures to make the story more accessible to children. This results in a network with fewer nodes and edges, reflecting a streamlined narrative and weakened interconnectedness. However, in *Through the Looking Glass*, the narrative structure is more complex and interconnected. Characters not only interact with Alice but also with each other, creating a denser and more elaborate network.

Character Dynamics In each book, Alice's role as the protagonist creates a strong central node with numerous direct connections to other characters. In *Alice's Adventures in Wonderland*, the network is complex, with a high character count and diverse interactions suitable for an engaging narrative. In *Nursery Alice*, Alice's centrality is even more pronounced, with fewer peripheral characters, highlighting key interactions in a simplified, linear network. In *Through the Looking Glass*, the network increases in complexity and density, reflecting Alice's navigation through a world with richer and more varied relationships.

Impact of Removing Characters

In Moretti's paper on Network Theory (Moretti [2011]), he mentions that the protagonist's significance or function is tied to the network's stability, which is closely related to centrality. In *Alice's Adventures in Wonderland*, removing Alice still leaves some interconnectedness among other characters, with certain relationships strengthening as they adapt to her absence. In *Nursery Alice*, removing Alice has a more noticeable impact due to the already simplified network. Each remaining character's role becomes more pronounced, further highlighting the importance of Alice's presence. This is expanded upon in the below 'Close-Reading Response and Network Perception' section.

Moretti also suggests considering the impact of removing the second most central character. Key characters that are structurally important affect the network significantly when removed. In *Alice's Adventures in Wonderland*, removing Pigeon (who is only qualitatively

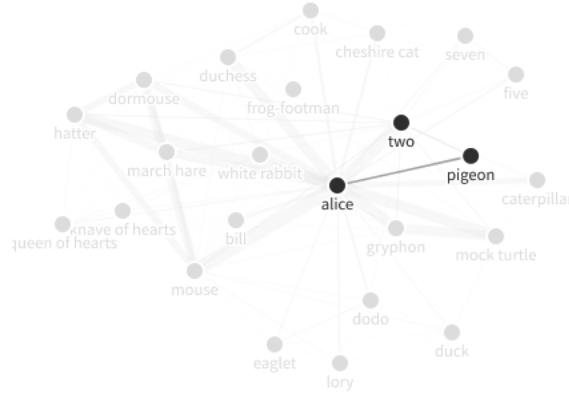


Figure 10: Pigeon's Interactions within *Alice's Adventures in Wonderland*

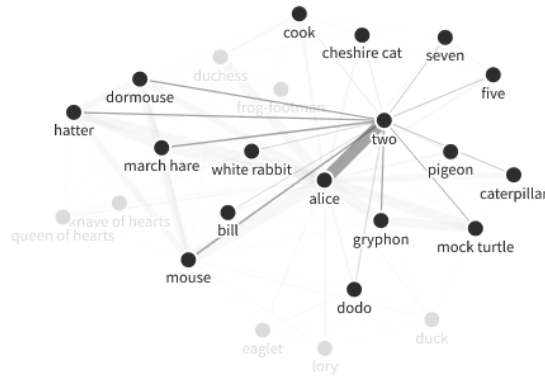


Figure 11: Two's Interactions within *Alice's Adventures in Wonderland*

significant) has little effect on the network due to limited interactions with other characters. In Figure 10 we can see Pigeon's interactions in the overall network. However, removing Two (structurally significant) disrupts the network's substructures, revealing the absence of its "world" within the overall network. In Figure 11 we can see Two's relationships within the overall network.

In *Through the Looking Glass*, removing Lily, an important structural character, visually weakens the interconnectedness between other characters, similar to the effect of removing Two in *Alice's Adventures in Wonderland*.

Moretti also highlights that once a network model is created, it abstracts the text to characters and interactions, revealing underlying structures otherwise hidden by the richness of the play. This reduction allows us to see certain regions within the narrative. While literary theory often focuses on concepts of "consciousness" and "interiority," network theory abstracts characters to their connections, placing the protagonist at the center of the network. This approach is why we included a close reading analysis to complement the network theory.

4.3 Close-Reading Response and Network Perception

In order to more fully understand what these networks mean in the context of this series, we are able to use our close-reading of the texts in combination with this project’s distant-reading network analysis. As authors Birkholz and Budke write,

”both approaches of reading [are] not at odds, but rather... when both methods are used in dialogue with one another, they allow us to fill gaps in our knowledge that we could otherwise not fill.” (Birkholz and Budke [2021]). As a research team, we collectively close-read all three of the novels in the Alice series. Throughout this reading, we noted character interactions, their positivity or negativity, and our general impressions.

With our observations from this close-reading, we can see that these networks indeed reflect the complexity we perceived about the character relationships of each novel.

In our close-reading, we noted that the difference in complexity of character relationships jumped dramatically between *The Nursery Alice* and *Alice’s Adventures in Wonderland* and this is reflected in their respective network visualizations. In our close-reading of *The Nursery Alice* specifically, we noted that there were far fewer characters in general and there seemed to be a greater emphasis on Alice as the sole main character. This may support younger readers comprehension of the novel as they are able to keep track of the limited amount of characters and imagine themselves as the sole main character, Alice, being a part of each important interaction. This is reflected in the network for *The Nursery Alice* as fewer characters appear compared to the network for *Alice’s Adventures in Wonderland* and Alice is the center-most character in *The Nursery Alice* network. No other characters in *The Nursery Alice*’s network come close to the centrality that Alice herself has, whereas, in the network for *Alice’s Adventures in Wonderland*, other characters such as Mouse, Two, Duchess, and Frog Footman also have many strong connections in the network and decent centrality. This shows that there is an increased complexity or difficulty in *Alice’s Adventures in Wonderland* when determining who the main character or main focus is. Older readers, the intended audience for *Alice’s Adventures in Wonderland*, therefore, may be able to handle and follow along with the larger cast of main characters and their varied amounts of interactions with others.

This idea is further supported when exploring one of the particular capabilities of this distant-reading network approach: the ability to remove a character from the network and see what remains.

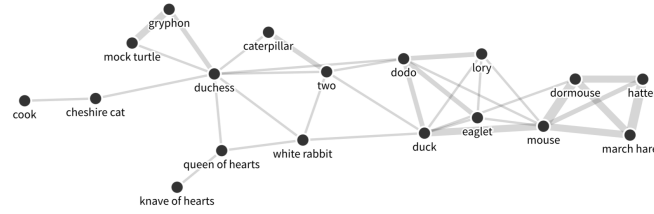


Figure 12: Flourish Network Visualization for *The Nursery Alice* when Alice is removed

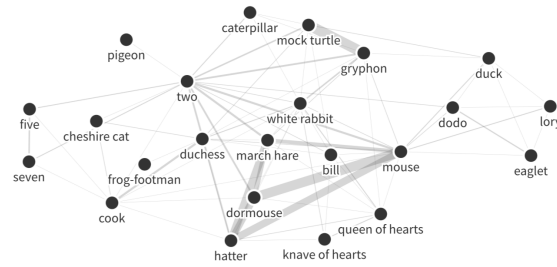


Figure 13: Flourish Network Visualization for *Alice's Adventures in Wonderland* when Alice is removed

Here we can see that the network of *The Nursery Alice* heavily relies on Alice as the central, main character to provide its structure. Many of the interactions of the novel hinge on her presence. The network completely flattens and loses shape and stability without Alice. On the other hand, when Alice is removed from *Alice's Adventures in Wonderland*, the network remains a cloud-like and rounded shape showing that Alice is less essential to maintaining the overall structure of the novel given the presence of other key main characters. *The Nursery Alice* therefore does more centrally use the character of Alice as a tool to guide the young readers through the simplified novel.

We also noted in our close-reading of *Alice's Adventures in Wonderland*, that Alice seemed to have a gradation of closeness with the other characters (i.e. a wide range of numbers of interactions between Alice and other characters), whereas *The Nursery Alice* had a more extreme differentiation between the closeness of characters (i.e. characters with either large numbers of interactions with Alice or very low numbers of interactions with Alice). We presume this choice may help very young readers understand which are Alice's most crucial relationships. This close-reading observation is reflected in the respective network visualizations through the thickness of the edges between character nodes. Alice's relationships in the *Alice's Adventures in Wonderland* network are of strong, medium-, or low-strength as compared to the only strong or weak relationships of *The Nursery Alice*

network. This can be seen in the gradation in thickness of edges in figure 5 as compared to the only thick or thin edges of figure 4. For example, the Mock Turtle has a weak connection with Alice in the network for *The Nursery Alice* but a medium-strength connection to Alice in the network for *Alice's Adventures in Wonderland*. This gradation of relationship strengths for characters in *Alice's Adventures in Wonderland* is a more complex structure than the simple, strong versus weak connection dynamic for the characters of *The Nursery Alice*.

Our subjective (but valid) feelings from our close-reading are therefore enriched and confirmed with our more objective distant-reading network analysis. This provides a stronger argument that *The Nursery Alice* indeed offers a simplified character relationship scheme than that of *Alice's Adventures in Wonderland*.

It is worth noting that the network for *Alice's Adventures in Wonderland* is entirely missing some characters. The absence of these characters may be due to the particular criteria used to calculate relationship scores and subsequently build this network. Characters whose names only occur a small handful of times in the novel may not have appeared in close proximity enough to any other character to be included in the network. (For example, "Ada" appears by name only three times in all of *Alice's Adventures in Wonderland*). This may have effected the shape and layout of the network and limit full comparisons between networks or novels and thus is an instance where distant-reading alone does not adequately express the ideas of the original text.

These inadequacies are also important to keep in mind when comparing the networks for *Alice's Adventures in Wonderland* and *Through the Looking-Glass, And What Alice Found There*. Here, we observe that the networks have fairly similar complexities visually. In this case, we can more deeply explore particular relationships within the network and novel. In the two networks we can see that Alice has significant interactions with the figures of authority around her. For example, Alice has a strong connection with the Duchess and a minor connection with the Queen of Hearts in the *Alice's Adventures in Wonderland* network. In the network for *Through the Looking-Glass, And What Alice Found There*, Alice has many more interactions with royalty and authority figures such as Red Queen, Red King, White Queen, White King, and White Knight. From these networks alone, we may be able to imply that Alice's role in *Through the Looking-Glass, And What Alice Found There* has increased in importance and maturity as she has more interactions with varying strengths with these leaders and royals.

We can then use our close-reading of the two texts to see how these increased interactions with royals may show a growth in the character of Alice matching the growth of the readers consuming the *Through the Looking-Glass, And What Alice Found There* novel. While the exact same characters do not appear in the two books, we can still compare Alice's interaction with a king in each of the novels. In *Alice's Adventures in Wonderland*, the King calls Alice a "child":

The King laid his hand upon her arm, and timidly said “Consider, my dear: she is only a child!”

This king in *Alice’s Adventures in Wonderland* never calls her a lady, woman, or any other term. On the other hand, the King in *Through the Looking-Glass, And What Alice Found There*, calls Alice “a young lady”:

“This young lady loves you with an H,” the King said, introducing Alice in the hope of turning off the Messenger’s attention from himself

This change in language could be accounted for by a change of character (these being two different kings) or be coincidence in the writing process, or it may point to underlying changes in Alice’s interactions with those around her including figures of authority.

While the network visualizations do reveal and support some insights about the texts, they cannot express the full complexity of the story and the types of interactions that occur. For example, in *Through the Looking-Glass, And What Alice Found There*, we noted from our close-reading that when compared to Alice’s interactions in *Alice’s Adventures in Wonderland*, Alice took on more mature roles in interactions with other characters. For example, she talks “in a soothing tone,” to characters around her and interacts “gently” with them:

“Never mind!” Alice said in a soothing tone, and stooping down to the daisies, who were just beginning again, she whispered, “If you don’t hold your tongues, I’ll pick you!”

Alice laid her hand upon his arm, and said in a soothing tone, “You needn’t be so angry about an old rattle.”

So Alice picked him up very gently, and lifted him across more slowly than she had lifted the Queen, that she mightn’t take his breath away: but, before she put him on the table, she thought she might as well dust him a little, he was so covered with ashes.

Here, Alice appears as more mature and caring for other characters she interacts with. These types of interactions cannot be shown through network visualizations and the full shift in complexity between *Alice’s Adventures in Wonderland* and *Through the Looking-Glass, And What Alice Found There* cannot be seen. Thus, in these instances, a combination of close and distant reading is needed.

5 Future Work

This project has potential for expansion and future work. One area that would enrich the connections displayed in the networks is co-reference resolution. Often pronouns are used in

place of character names. For example, in this passage extracted from *Alice’s Adventures in Wonderland*, “she” refers to Alice, but with our current capabilities, the connection between Alice and the White Rabbit would not be captured:

But **she** went on all the same, shedding gallons of tears, until there was a large pool all round her, about four inches deep and reaching half down the hall.

After a time **she** heard a little pattering of feet in the distance, and **she** hastily dried her eyes to see what was coming. It was the **White Rabbit** returning, splendidly dressed, with a pair of white kid gloves in one hand and a large fan in the other: he came trotting along in a great hurry, muttering to himself as he came, “Oh! the Duchess, the Duchess! Oh! won’t she be savage if I’ve kept her waiting!”

In this quote, the actual name *Alice* does not appear within two lines of the *White Rabbit*, so it will not be captured in our current relationship score system. However, if co-reference resolution were undertaken, a previous occurrence of *Alice* would be connected to the above occurrences of *she* which would then be connected to the *White Rabbit* and thus, the connection between Alice and the White Rabbit would be enriched and more accurate to the events of the novel.

6 Limitations

Limitations of our current study include the size of our data inputs. The short Alice series of only three novels (that are short enough in length to be suitable for children) do not offer very large amounts of material to perform our analysis on. Additionally, we are constrained methodologically to the network visualization tools available to us. We would not be able to build and customize our own package to perform this step due to time and coding experience constraints nor can we access network tools that are held by a non-open source license. Lastly, as we did not receive individual written feedback following our group’s presentation, we held our focus on the oral feedback given that presentation day and pursued our best attempt on how to move forward with this project.

7 Conclusion

Through this network analysis exploration of the *Alice in Wonderland* series, we have shown that the complexity of the relationships indeed increases across the three novels. This distant-reading analysis, which leverages quantitative methods to uncover patterns and structures in the text, is complemented when combined with close-reading techniques. Close reading allowed us to track not only the increase in complexity over time but also the specific ways in which character dynamics developed. For instance, we could analyze

Alice's interactions with royalty and figures of authority in more detail, observing how these relationships evolved and contributed to the narrative progression.

In this project, we also compared different tools for network building and visualization. Our findings suggest that the choice of tools depends largely on the objective of the analysis. NetworkX, with its simplicity and ease of learning, is well-suited for users who require straightforward and rapid visualization capabilities. In contrast, Gephi offers a highly customizable and advanced platform, making it ideal for more complex and detailed network analysis. These tools complement each other, providing a range of options for researchers depending on their specific needs and expertise.

Our exploration using the Alice series emphasizes the potential of network theory and the accompanying analysis techniques to provide valuable insights into a wide range of texts, including novels, plays, and other literary forms. Network analysis extends traditional literary analysis by offering a different perspective, allowing researchers to back up certain interpretations or challenge them with quantitative data. This powerful tool can reveal underlying patterns and connections that might not be immediately apparent through close reading alone.

Additionally, as discussed in the 'Future Work' section 5, there are opportunities to extend our work. By providing relationship scores that account for instances where characters are referred to by pronouns or other inferred means, we may be able to reveal richer and more nuanced depictions of these character relationships. With this area for improvement in mind, we are still able to confirm that our project provides valuable insights into the character relationships of these novels and their development in complexity over the course of the series.

Acknowledgements

This project was completed as a part of the curriculum for the University of Zürich course: "Text Analytics in the Digital Humanities" offered in the Spring Semester 2024 under the teaching of Prof. Dr. Gerold Schneider and tutor Ahmet Uluslu. Chat-GPT was used sparingly during this project to provide support to fix coding bugs and when creating this PDF on Overleaf (most often when creating tables and the other formatting elements of this paper). This project was possible thanks to the availability of Lewis Carroll's ever-influential "Alice" works.

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